FY121A Circuit Engineering
Lab Report – Part 4



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Course Title:	FY121A Circuit Engineering
Assignment:	Lab 4
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Task 1

1. Theoretical Part

Since the voltmeter measures the voltages in RMS, the amplitude of the input can be changed to the RMS value by dividing it with $\sqrt{2}$:

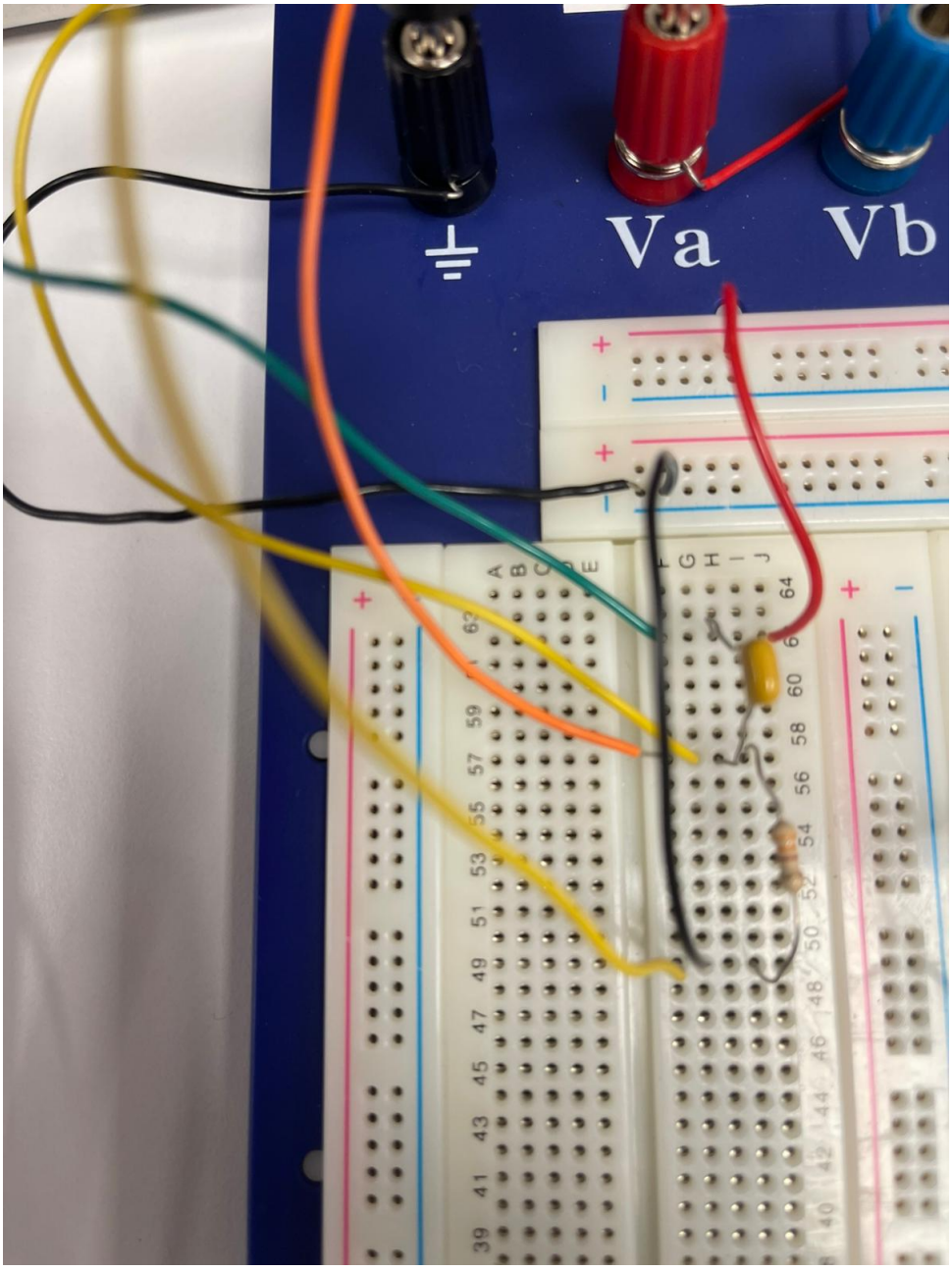


Figure 1: circuit design task 1

$$V_{in,rms} = \frac{\hat{V}}{\sqrt{2}} = \frac{2.0}{\sqrt{2}} \approx 1.414\,\mathrm{V}$$

1. Calculations at $f = 200$ Hz

In order to set up a voltage divider to determine the respective voltages for X_C and R, both X_C and the absolute value of the total impedance $|Z|$ must be calculated:

$$X_C = \frac{1}{2\pi(200)(1 \times 10^{-6})} \approx 795.77\,\Omega$$
$$|Z| = \sqrt{330^2 + 795.77^2} \approx 861.54\,\Omega$$

With X_C and $|Z|$ calculated, a voltage divider can now be used to determine V_C and V_R:

$$V_{R,rms} = 1.414 \cdot \frac{330}{861.54} \approx 0.541\,\text{V}$$
$$V_{C,rms} = 1.414 \cdot \frac{795.77}{861.54} \approx 1.306\,\text{V}$$

The phase shift (ϕ) can then be determined by the by taking the arctan() of X_C over R

$$\phi = \arctan\left(\frac{795.77}{330}\right) \approx 67.48^\circ$$

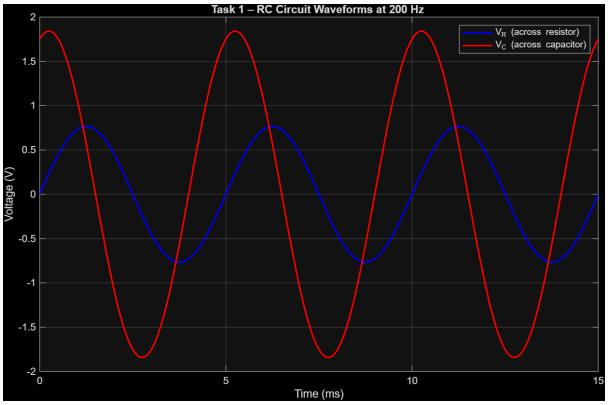


Figure 2: theoretical Graph for 200Hz

2. Calculations at $f = 1$ kHz

To find the values at $1000Hz$, the previous calculations can be repeated, but with the new f value:

$$X_C = \frac{1}{2\pi(1000)(1 \times 10^{-6})} \approx 159.15\,\Omega$$
$$|Z| = \sqrt{330^2 + 159.15^2} \approx 366.38\,\Omega$$

The resulting RMS voltages are:

$$V_{R,rms} = 1.414 \cdot \frac{330}{366.38} \approx 1.274\,\text{V}$$
$$V_{C,rms} = 1.414 \cdot \frac{159.15}{366.38} \approx 0.614\,\text{V}$$

The theoretical phase shift ϕ is:

$$\phi = \arctan\left(\frac{159.15}{330}\right) \approx 25.75^\circ$$

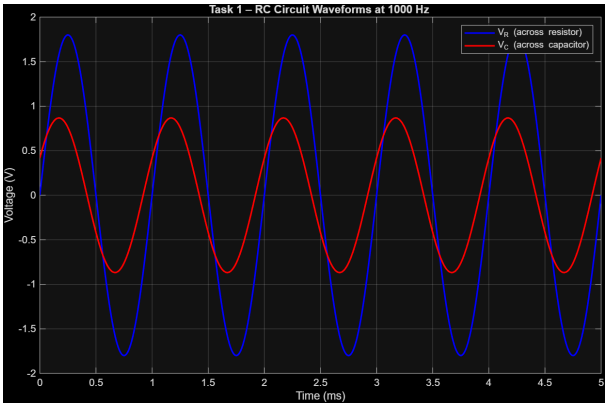


Figure 3: THeoretical Graph for 1KHz

2. Measurement Values

Figure 4: Oscilloscope image at 200 Hz

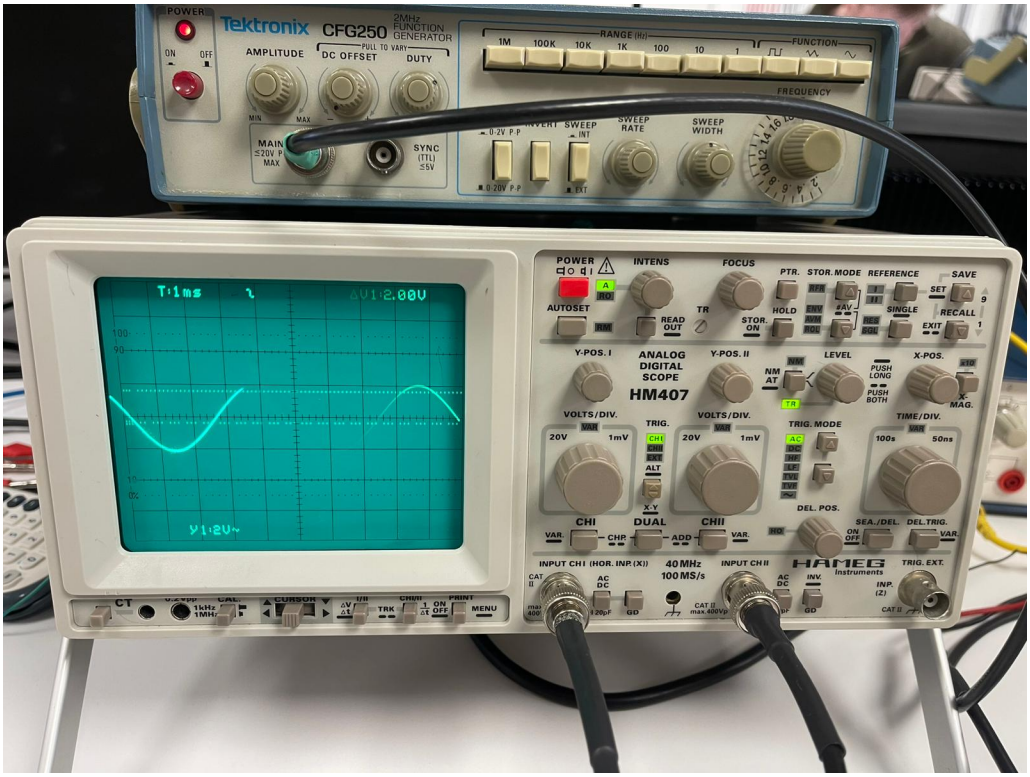


Figure 4: Oscilloscope measurement at 200 Hz.

Figure 5: Oscilloscope image at 1 kHz

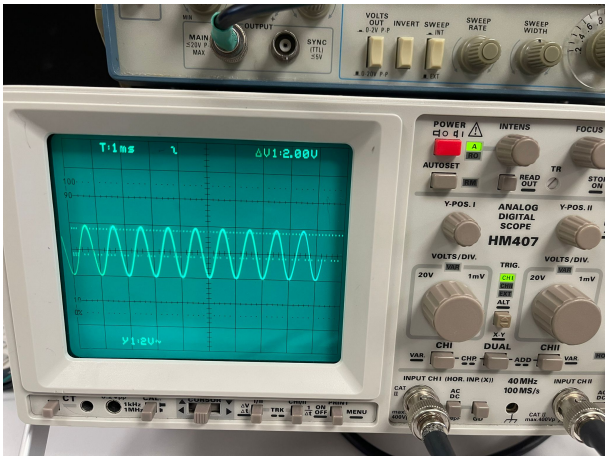


Figure 5: Oscilloscope measurement at 1 kHz.

Table 1: Measured Voltage and Phase Shift Values

Frequency (Hz)	$V_{R,rms}$ (V)	$V_{C,rms}$ (V)	**Measured** ϕ (°)
200	0.497	1.365	72.00
1000	1.136	0.655	28.80

3. Comparison With Theory

Table 2: Comparison of RMS Measured and Theoretical Voltage Values

	$V_{R,rms}$ **(V)**			$V_{C,rms}$ **(V)**		
Freq. (Hz)	**Meas.**	**Theo.**	**% Dev.**	**Meas.**	**Theo.**	**% Dev**
200	0.497	0.541	8.13	1.365	1.306	4.52
1000	1.136	1.274	10.83	0.655	0.614	6.68

Table 3: Comparison of Measured and Theoretical Phase Shift (ϕ)

Frequency (Hz)	**Measured** ϕ (°)	**Theoretical** ϕ (°)	**% Dev.**
200	72.00	67.48	6.69
1000	28.80	25.75	11.84

4. Conclusions

Reactance, in the form of X_c, was observed to inversely relate to the level of frequency. As can be seen in Tables 1 and 2, the voltage over the capacitor at $200Hz$ was measured at $1.365V$, but subsequently dropped to $0.655V$ once the frequency was changed to $1000Hz$. Conversely, the voltage over the resistor went from a measured voltage of $0.497V$ at $200Hz$, to $1.136V$

at $1000Hz$. This relationship between the frequency and voltage over the capacitor suggests that as the frequency of the input rises, X_c actually decreases. Further verification of this can be seen when looking at the measured phase shift values in Tables 1 and 3. Due to reactance and resistance having an orthogonal relationship, the phase shift value describes the degree of reactance over resistance. The observed change from a measured ϕ of 72° at $200Hz$ to one of 28.8° at $1000Hz$ demonstrates a shift toward the resistive axis on the impedance plane.

Task 2

1. Theoretical Part

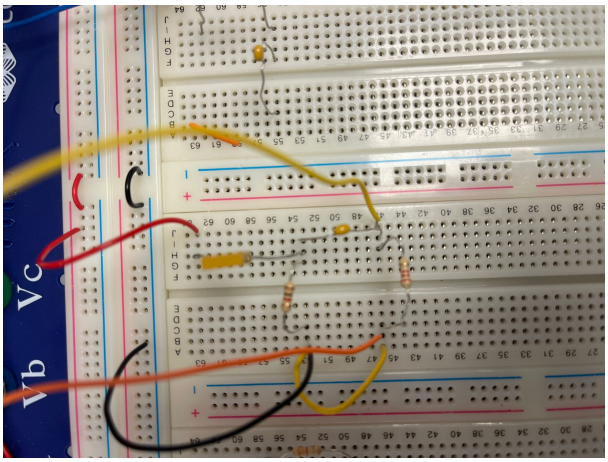


Figure 6: Circuit Design for Task2

The first step in calculating the current over R_2 is to calculate ω, which can then be used to determine the impedance values X_{C_1} and X_{C_2}:

$$\omega = 2\pi f = 2\pi(1000) \approx 6283.19\,\text{rad/s}$$
$$X_{C_1} = \frac{1}{j\omega C_1} = -j\frac{1}{6283.19 \cdot 0.47 \times 10^{-6}} \approx -j338.6\,\Omega$$
$$X_{C_2} = \frac{1}{j\omega C_2} = -j\frac{1}{6283.19 \cdot 0.1 \times 10^{-6}} \approx -j1591.5\,\Omega$$

Then, in order to set up a voltage divider equation, X_{C_1} and X_{C_2} need to be combined:

$$Z_{branch} = Z_{C_2} + R_2 = 2200 - j1591.5\,\Omega$$

Followed by combining R_1 with Z_{branch}, which are parallel to one another:

$$Z_{branch} = R_2 + X_{C_2} = 2200 - j1591.5 \approx 2715.3\angle -35.90^\circ\,\Omega$$
$$Z_p = \frac{R_1 \cdot Z_{branch}}{R_1 + Z_{branch}} = \frac{1000 \cdot (2200 - j1591.5)}{3200 - j1591.5}$$
$$Z_p \approx 694.7 - j152.1 \approx 711.2\angle -12.35^\circ\,\Omega$$

With the circuit now condensed, a voltage divider can be used to find the voltage that goes through Z_p, which is the same voltage that goes through Z_{branch}:

$$V_{branch} = V_{in}\frac{Z_p}{X_{C_1} + Z_p}$$
$$V_{branch} = 6.0\angle 0^\circ \frac{694.7 - j152.1}{-j338.6 + (694.7 - j152.1)}$$
$$V_{branch} = 6.0\angle 0^\circ \frac{711.2\angle -12.35^\circ}{694.7 - j490.7}$$
$$V_{branch} \approx 6.0\angle 0^\circ \frac{711.2\angle -12.35^\circ}{850.5\angle -35.24^\circ} \approx 5.02\angle 22.89^\circ\,\text{V}$$

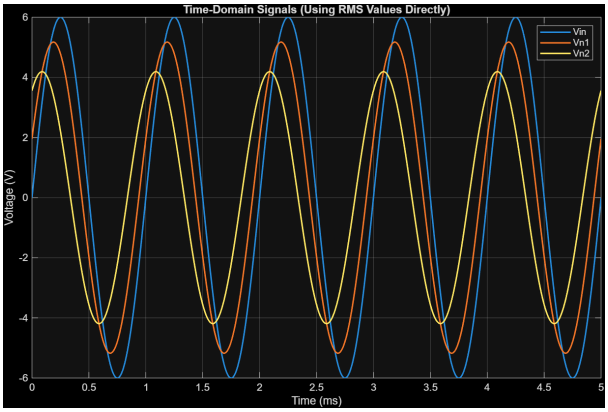


Figure 7: Theoretical graph

Another voltage divider will then give the voltage over R_2 the phase shift (ϕ):

$$
\begin{aligned}
V_{R_2} &= V_{branch} \frac{R_2}{X_{C_2} + R_2} \\
V_{R_2} &= 5.02\angle 22.89^\circ \frac{2200}{2200 - j1591.5} \\
V_{R_2} &= 5.02\angle 22.89^\circ \frac{2200}{2715.3\angle -35.90^\circ} \\
V_{R_2} &\approx 4.07\angle 58.79^\circ \, \mathbf{V}
\end{aligned}
$$

Finally, dividing V_{R_2} by R_2 gives the value of the current I_{R_2}:

$$
\begin{aligned}
I_{R_2} &= \frac{V_{R_2}}{R_2} = \frac{4.07\angle 58.79^\circ}{2200} \\
I_{R_2} &\approx 1.85\angle 58.79^\circ \, \mathbf{mA}
\end{aligned}
$$

2. Measurement Values

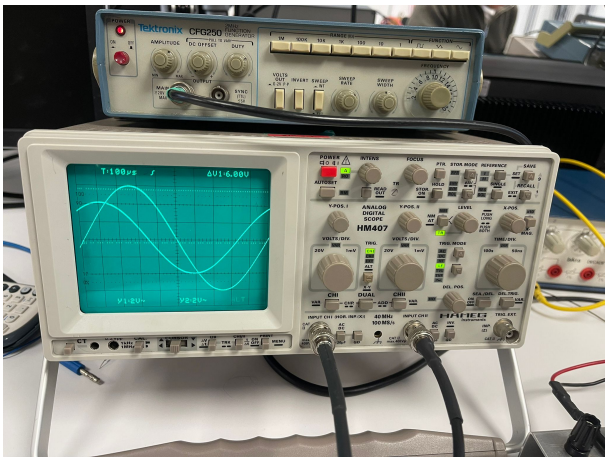


Figure 8: Practical Graph

Figure 9: Oscilloscope measurement for Task 2.

Table 4: Measured Values for Task 2

Parameter	**Symbol**	**Measured Value**
Input Amplitude (Reference)	$\hat{V}_{in}$	6.0 V
Voltage over R_2	$\hat{V}_{R_2}$	4.05 V
Phase Shift (relative to V_{in})	ϕ	54.0°

3. Comparison With Theory

Table 5: Comparison of Theoretical and Measured Results

Quantity	**Theoretical**	**Measured/Derived**	**Error (%)**
Voltage $\hat{V}_{R_2}$ (V)	4.07	4.05	0.49%
Current $\hat{I}_{R_2}$ (mA)	1.85	1.84	0.54%
Phase ϕ (deg)	58.79	54.0	8.87%

4. Conclusions

Given how closely the measured values approached the theoretical values, it shows how accurately calculations done on the complex plane portray the relationship between impedance (Z) and alternating currents. When compared to DC circuits, AC introduces the concept of reactance (X) in addition to resistance (R), which necessitates these complex calculations.

Also, the $+54.0^{\circ}$ phase shift also tells us that the current is leading the voltage in this instance. Given that this is the case with capacitors, this makes sense. With this phase shift, the peak output arrives approximately $0.15ms$ before the peak input.

Task 3

1. Theoretical Part

To calculate the cutoff frequency f_c, we take the inverse value of $2\pi RC$. For the magnitude A_v and the phase shift ϕ, they can be derived as seen below:

$$f_c = \frac{1}{2\pi R_1 C_1} \approx 1591.55 \text{ Hz}$$

$$A_v(f) = \left| \frac{1}{1 + j2\pi f R_1 C_1} \right| = \frac{1}{\sqrt{1 + (2\pi f R_1 C_1)^2}} = \frac{1}{\sqrt{1 + (f/f_c)^2}}$$

$$\phi(f) = \arg\left(\frac{1}{1 + j\frac{f}{f_c}} \right) = -\arctan\left(\frac{f}{f_c} \right)$$

Theoretical Calculations for $100Hz$

$$A_v = \frac{1}{\sqrt{1 + (100/1591.55)^2}} \approx 0.998$$
$$\phi = -\arctan\left(\frac{100}{1591.55} \right) \approx -3.60^\circ$$

Theoretical Calculations for Measured Cutoff of $1538.46Hz$

$$A_v = \frac{1}{\sqrt{1 + (1538.46/1591.55)^2}} \approx 0.719$$
$$\phi = -\arctan\left(\frac{1538.46}{1591.55} \right) \approx -44.03^\circ$$

Theoretical Cutoff of $1591.55Hz$

$$A_v = \frac{1}{\sqrt{1 + (1591.55/1591.55)^2}} = \frac{1}{\sqrt{1 + 1^2}} = \frac{1}{\sqrt{2}} \approx 0.707$$

$$\phi = -\arctan\left(\frac{1591.55}{1591.55} \right) = -\arctan(1) = -45.00^\circ$$

Theoretical Calculations for $20kHz$

$$A_v = \frac{1}{\sqrt{1 + (20000/1591.55)^2}} \approx 0.079$$
$$\phi = -\arctan\left(\frac{20000}{1591.55} \right) \approx -85.45^\circ$$

2. Measurement Values

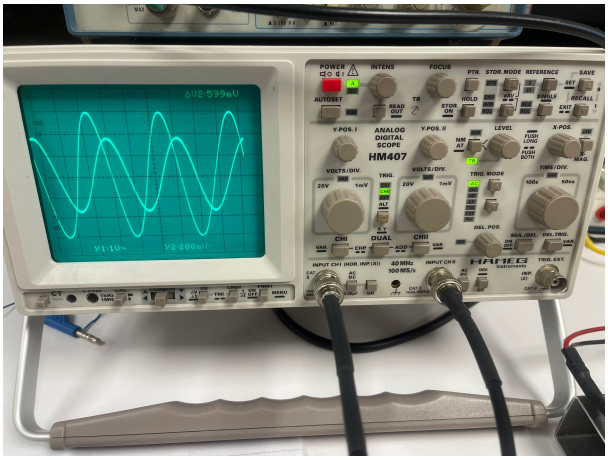
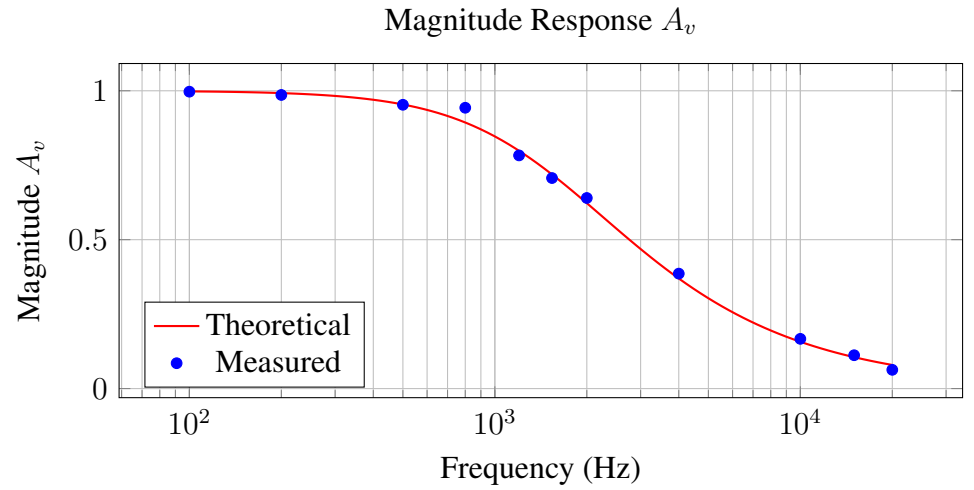
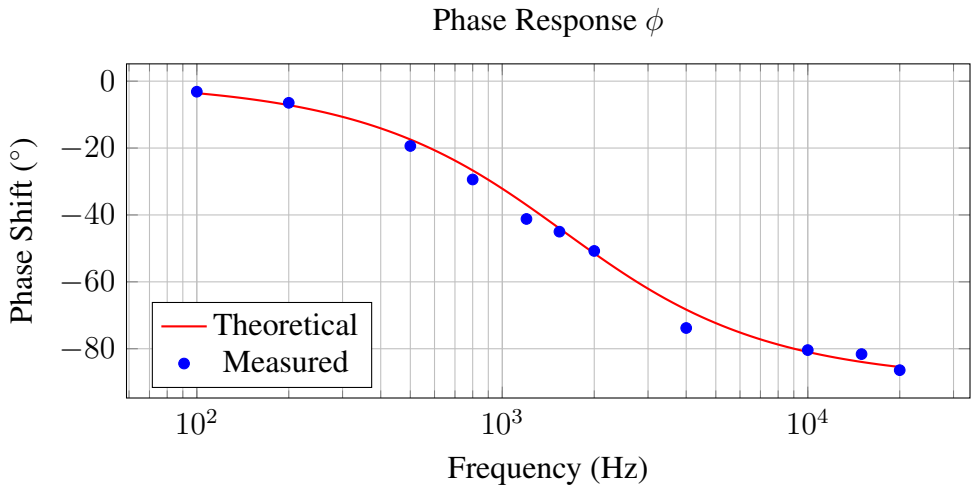


Figure 10: Measurement at 15kHz

Table 6: Measured Magnitudes and Phase Shifts ($V_{in} = 3.0$ V)

Frequency (Hz)	**Measured Magnitude** A_v	**Measured Phase** ϕ (°)
100	0.997	-2.16
200	0.986	-6.48
500	0.953	-19.38
800	0.943	-29.40
1200	0.783	-41.20
1538.46	0.707	-45.00
2000	0.640	-50.76
4000	0.386	-73.80
10000	0.167	-80.40
15000	0.112	-81.60
20000	0.063	-86.40





3. Comparison With Theory

Table 7: Magnitude and Phase Comparison ($V_{in} = 3.0$ V)

	Magnitude A_v			**Phase ϕ (°)**		
Freq (Hz)	**Theo.**	**Meas.**	**% Err**	**Theo.**	**Meas.**	**% Err**
100	0.998	0.997	0.10%	-3.60	-3.16	13.92%
200	0.992	0.986	0.60%	-7.20	-6.48	10.00%
500	0.954	0.953	0.10%	-17.44	-19.38	11.12%
800	0.893	0.943	5.60%	-26.70	-29.40	10.11%
1200	0.798	0.783	1.88%	-37.00	-41.20	11.35%
1538.46	**0.719**	**0.707**	**1.67%**	**-44.03**	**-45.00**	**2.20%**
2000	0.622	0.640	2.89%	-51.50	-50.76	1.44%
4000	0.369	0.386	4.61%	-68.30	-73.80	8.05%
10000	0.157	0.167	6.37%	-81.00	-80.40	0.74%
15000	0.105	0.112	6.67%	-83.90	-81.60	2.74%
20000	0.053	0.063	18.87%	-85.45	-86.40	1.11%

4. Conclusions

In the theoretical calculations (Table 7), when $f = 1.591.6$, X_C and R become equal in magnitude, creating a 45° phase shift from their total impedance. In the lab, f_c was measured to be a slightly lower value, which would most likely be explained by the tolerance of the capacitor and resistor used, which should be expected to vary in some degree from their advertised values. The significance of this f_c value is that it is when the power output is halved. Looking at the A_v value of .707, this is equal to $\frac{1}{\sqrt{2}}$, which when squared provides the power ratio $\frac{1}{2}$.

This task also further described the relationship between capacitance and frequency. When frequencies are low, X_C is high, resulting in a higher V_{out}. As frequencies rise, X_C lowers, siphoning voltage from the top node, causing V_{out} to diminish.

Task 4

1. Theoretical Part

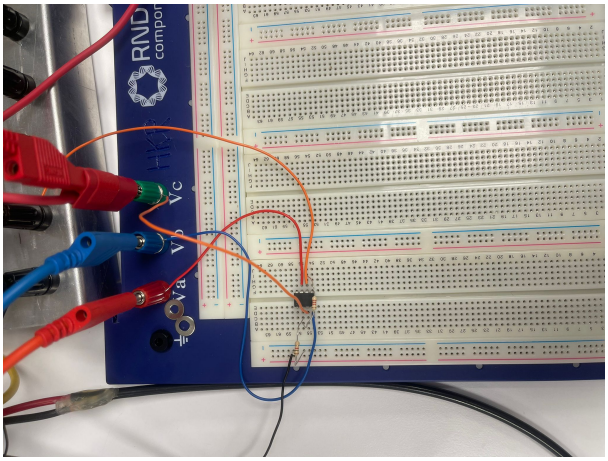


Figure 11: Circuit for Lab 4

OP-AMP Gain (A_0):

$$A_0 = 1 + \frac{R_2}{R_1} \tag{1}$$

For $R_2 = 199\text{k}\Omega$ and $R_1 = 1\text{k}\Omega$:

$$A_{0_1} = 1 + \frac{199\text{k}\Omega}{1\text{k}\Omega} = 1 + 199 = 200$$

For $R_2 = 399\text{k}\Omega$ and $R_1 = 1\text{k}\Omega$:

$$A_{0_2} = 1 + \frac{399\text{k}\Omega}{1\text{k}\Omega} = 1 + 399 = 400$$

Cut-off frequency (f_c):

$$f_c = \frac{GBP}{A_0} \tag{2}$$

For $A_{0_1} = 200$:

$$f_{c,A} = \frac{1 \cdot 10^6 \text{ Hz}}{200} = 5,000 \text{ Hz} = 5 \text{ kHz}$$

For $A_{0_2} = 400$:

$$f_{c,B} = \frac{1 \cdot 10^6 \text{ Hz}}{400} = 2,500 \text{ Hz} = 2.5 \text{ kHz}$$

The A_v value at any given frequency can be found using the following formula, which was derived previously in Task 3:

$$A_v(f) = \frac{A_0}{\sqrt{1 + \left(\frac{f}{f_c}\right)^2}}$$

Once the cutoff frequency is reached ($f = f_c$), A_v should drop to:

$$A_v(f_c) = \frac{A_0}{\sqrt{1+1^2}} = \frac{A_0}{\sqrt{2}} \approx 0.707 \cdot A_0$$

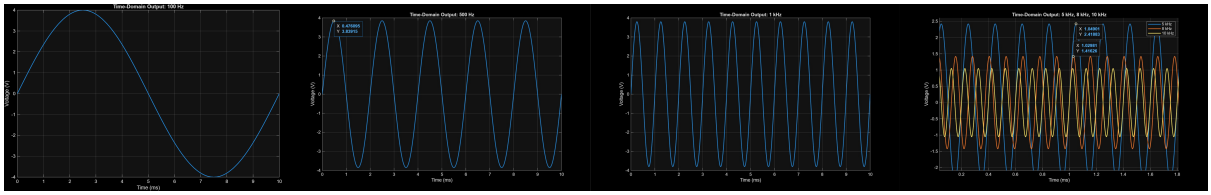


Figure 12: Graph for 199KHz

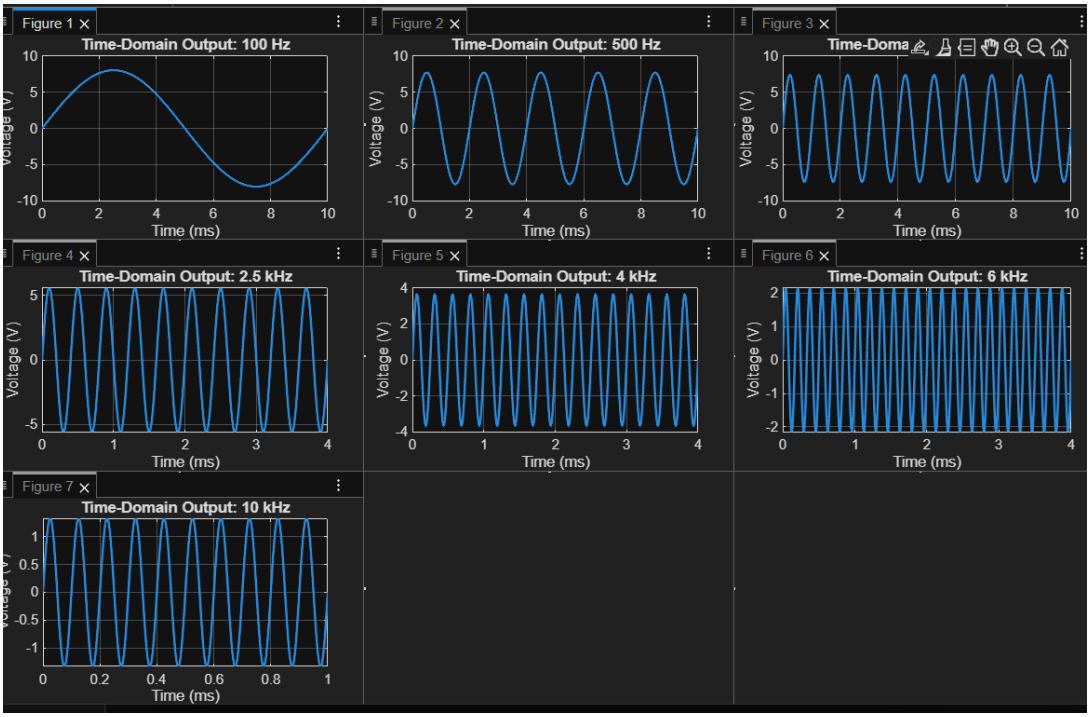


Figure 13: Graph for 399KHz

2. Measurement Values

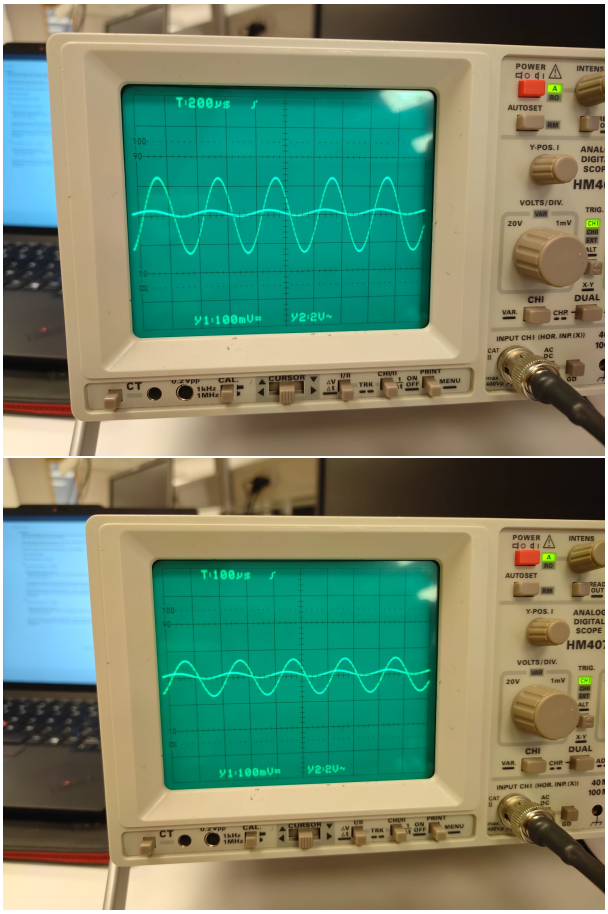


Figure 14: Oscilloscope measurement for Task 4.

Table 8: Measured Data for $R_2 = 199\text{k}\Omega$

Freq (Hz)	V_{in} **(mV)**	V_{out} **(V)**	**Measured** A_v
100	20	4.00	200.0
500	20	3.95	197.5
1000	20	3.86	193.0
5000	20	2.26	113.0
6000	20	1.92	96.0
8000	20	1.41	70.5
10000	20	1.05	52.5

Table 9: Measured Data for $R_2 = 399\text{k}\Omega$

Freq (Hz)	V_{in} **(mV)**	V_{out} **(V)**	**Measured** A_v
100	20	8.06	403.0
500	20	7.72	386.0
1000	20	7.38	369.0
2500	20	5.57	278.5
4000	20	3.65	182.5
6000	20	2.30	115.0
10000	20	1.13	56.5

3. Comparison With Theory

Table 10: Theoretical vs Measured Values - $R_2 = 199k$

Freq (Hz)	**Theo.** A_v	**Meas.** A_v	**% Error**
100	200.0	200.0	0.00%
500	199.0	197.5	0.75%
1000	196.1	193.0	1.58%
5000 (Cutoff)	**141.4**	**113.0**	**20.08%**
6000	128.0	96.0	25.00%
8000	106.0	70.5	33.49%
10000	89.4	52.5	41.28%
GBP (MHz)	**1.00**	**0.80***	**20.00%**

Table 11: Theoretical vs Measured Values - $R_2 = 399k$

Freq (Hz)	**Theo.** A_v	**Meas.** A_v	**% Error**
100	400.0	403.0	0.75%
500	392.2	386.0	1.58%
1000	371.4	369.0	0.65%
2500 (Cutoff)	**282.8**	**278.5**	**1.52%**
4000	212.1	182.5	13.96%
6000	153.8	115.0	25.23%
10000	97.0	56.5	41.75%
GBP (MHz)	**1.00**	**0.98***	**2.00%**

**Measured GBP calculated as* $A_{v(low_freq)} \times f_{c(measured)}$

4. Conclusions

The data suggests that frequency and gains in a closed-loop circuit have an inverse relationship. When the maximum gain A_0 is increased, the cutoff frequency f_c is decreased. According to

the theoretical calculations, given that $GBP = f_c \cdot A_0$, and GBP is a constant, doubling either A_v or the frequency bandwidth (represented by f_c), halves the other. To achieve the highest possible gains, there must be an equal sacrifice to the frequency bandwidth. Likewise, if one intends to allow for the maximum bandwidth, it must be with the knowledge that the gain will be at its minimum.

The increasing error percentages seen in Tables 10 and 11 suggest that the OP-AMP struggled to maintain expected gains as the frequencies rose. It is possible that this is due to the slew rate, which is where the rate of change in voltage presented by the input frequency becomes greater than the OP-AMP can keep pace with.

Task 5

1. Theoretical Part

Deriving the Slew Rate Equation:

Given that the slew rate is the maximum rate of change of an OP-AMP output voltage, the first step in deriving the formula is to start with the output signal:

$$V(t) = \hat{V}\cos(\omega t)$$

Then, the derivative is taken to find the slope of that signal:

$$\frac{d}{dt}V(t) = -\omega\hat{V}\sin(\omega t)$$

Taking the derivative once more, and setting it equal to zero, gives the point at which the rate of change of that slope reaches a maximum:

$$\frac{d^2}{dt^2}V(t) = -\omega^2\hat{V}\cos(\omega t) = 0 \implies \omega t = \frac{\pi}{2}$$

Having found when the value of t at which the slope is a maximum, we can substitute that value back into the first derivative (taking the magnitude) to find the maximum slope, which is the slew rate:

$$\text{Slew Rate} = |-\omega\hat{V}\sin(\pi/2)| = \omega\hat{V} = 2\pi f_m\hat{V}$$

This can also be re-arranged to give the formula for the maximum frequency f_m:

$$f_m = \frac{\text{Slew Rate}}{2\pi\hat{V}}$$

Calculating the Theoretical Values:

Closed-Loop Gain: $A_0 = 1 + \frac{2.2}{1} = 3.2$
Peak Output Voltage: $\hat{V}_{out} = 3.0\,\text{V} \cdot 3.2 = 9.6\,\text{V}$
Max Frequency: $f_m = \frac{500{,}000}{2\pi(9.6)} = 8,289\,\text{Hz}$
Half of f_m: $f = f_m/2 = 4145\,\text{Hz}$
Double f_m: $f = 2 \cdot f_m = 16578\,\text{Hz}$

2. Measurement Values

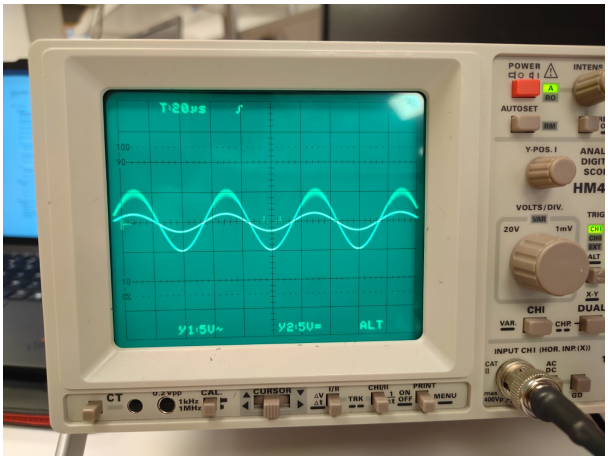


Figure 15: Oscilloscope measurement at $\frac{f_{max}}{2}$.

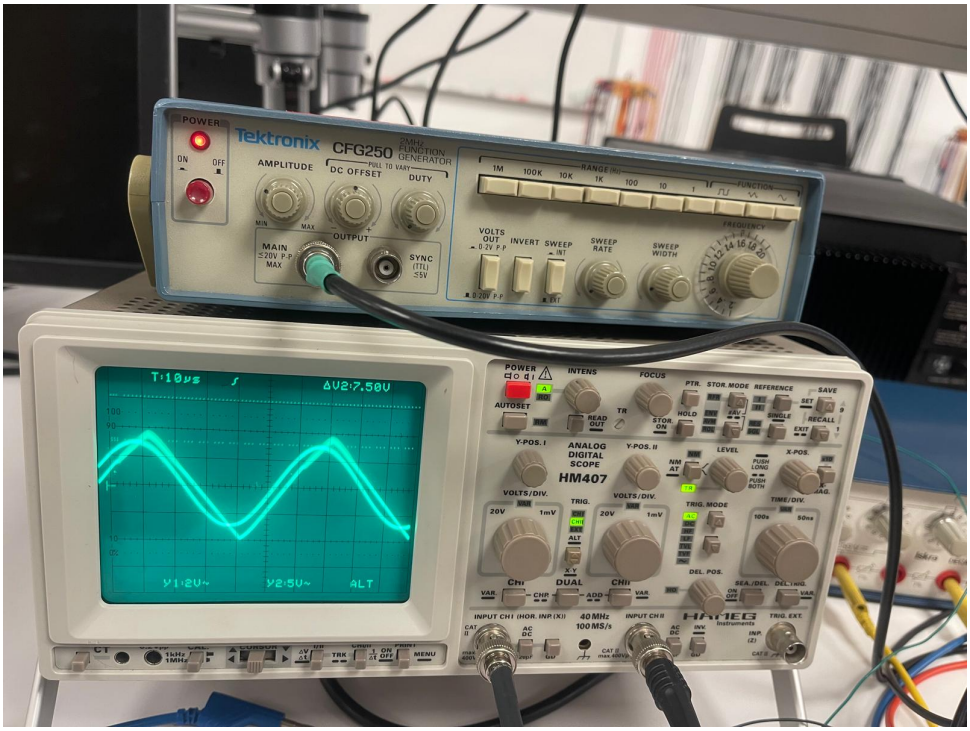


Figure 16: Oscilloscope measurement at $2 \cdot f_{max}$.

Table 12: Measured Voltages

Condition	**Freq (Hz)**	**Meas. $\hat{V}_{in}$ (V)**	**Meas. $\hat{V}_{out}$ (V)**
$f_{max}/2$	4145	3.0	8.77
$2 \cdot f_{max}$	16578	3.0	9.00

3. Comparison With Theory

Table 13: Theoretical vs Measured Values

Freq (Hz)	**Theo.** A_0	**Meas.** A_v	**% Error**
4145	3.2	2.92	8.75
16578	3.2	3.00	6.25

4. Conclusions

The slew rate should theoretically affect the measured output of the OP-AMP, but this was not seen in the lab measurements. Given the maximum slope of the OP-AMP output being half of what the input frequency would require, the output voltage should have only been able to reach approximately half of the peak before the voltage signal would drop again. However, the sine wave on the oscillator did appear closer to a triangular wave function, which does match with what should happen in theory. This triangular shape is a result of the slope of the input being too great, so the output slope stays a its constant maximum slope when rising and falling.

References

[1] FY121A Laboratory Session Part 4, Kristianstad University. [2] OP AMP 741 Datasheet. [3] Course Lecture Notes, FY121A.